



COMPUTER NETWORKS – SPRING 2026

Lab 3

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1. Overview

This project implements a multi-client TCP server using Java sockets. The system allows multiple clients to connect to the server at the same time, send commands, and receive processed responses. The server performs simple text operations such as converting case, reversing strings, counting characters, and retrieving the current date or time. The design focuses on simplicity, concurrency, and clear communication between client and server.

2. System Architecture

The system is composed of three main parts working together. The server listens for incoming connections on a fixed port and continuously accepts new clients. For every new connection, a separate thread is created to handle that client independently, which allows multiple clients to operate simultaneously without blocking each other. The client program connects to the server, sends user commands from the console, and displays the responses returned by the server.

3. Features and Commands

The server supports a set of predefined commands that follow a simple format. Some commands operate on text provided by the client, such as converting text to uppercase or lowercase, reversing a string, counting characters, or counting vowels. Other commands do not require input, such as retrieving the current system date or time. If a client sends an unknown command, the server responds with an error message indicating that the command is not recognized.



4. Implementation Details

The implementation relies on multithreading to allow concurrent client handling. Each client is managed by a dedicated thread, which ensures that communication remains responsive even when multiple users are connected. To prevent conflicts when multiple threads attempt to write to the same log file, synchronization is used so that only one thread can access the file at a time. The communication between client and server is implemented using buffered input streams for reading data and print writers for sending responses, which simplifies line-based communication.

The server also maintains a logging system that records all important events, including client connections, requests, responses, and disconnections. Each log entry includes both the current date and time to make tracking easier. This logging mechanism helps in debugging and monitoring system behavior.

5. Usage Instructions

To run the system, the Java files must first be compiled. After compilation, the server should be started, which will begin listening for incoming client connections. Once the server is running, one or more client programs can be launched. Each client connects to the server and allows the user to enter commands through the console. The server processes each command and sends back the result, which is then displayed to the user. The connection remains active until the user sends the quit command.

6. Dependencies

The project depends only on standard Java libraries, including networking, input/output handling, and date/time utilities. No external libraries or frameworks are required, which makes the project easy to run on any system with a Java Development Kit installed.



7. Design Considerations

The design follows a simple and modular approach to keep the system easy to understand and extend. Using a separate thread for each client ensures responsiveness and scalability for small to medium workloads. The command-based communication protocol makes it straightforward to add new features in the future.

Synchronization is used carefully to avoid data corruption without significantly affecting performance.



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Test Cases:

Client 1:

UPPER:hello world

COUNT:network

Client 2:

LOWER:PROGRAMMING

REVERSE:abcdef

Client 3:

VOWELS:aeiouxyz

COUNT:multithreading

Client 2:

TIME

QUIT

Client 1:

TIME

VOWELS:communication

Client 3:

DATE

UNKNOWN:test

Client 3:

QUIT

Client 1:

QUIT



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OUTPUTS:

SERVER:

```
NetLab2 (run) × NetLab2 (run) #2 × NetLab2 (run) #3 × NetLab2 (run) #4 ×
run:
Server waiting for clients
Client1 connected from 127.0.0.1:59037
Client2 connected from 127.0.0.1:59038
Client3 connected from 127.0.0.1:59039
Client1 Command: UPPER:hello world Result: HELLO WORLD
Client1 Command: COUNT:network Result: 7
Client2 Command: LOWER:PROGRAMMING Result: programming
Client2 Command: REVERSE:abcdef Result: fedcba
Client3 Command: VOWELS:aeiouxyz Result: 5
Client3 Command: COUNT:multithreading Result: 14
Client2 Command: TIME Result: 18:05:27.580497
Client 2 disconnected
Client1 Command: TIME Result: 18:05:44.678674
Client1 Command: VOWELS:communication Result: 6
Client3 Command: DATE Result: 2026-04-07
Client3 Command: UNKNOWN:test Result: ERROR: Unknown command from Client 3.
Client 3 disconnected
Client 1 disconnected
|
```

Client 1:

```
NetLab2 (run) × NetLab2 (run) #2 × NetLab2 (run) #3 × NetLab2 (run) #4 ×
run:
client is ready
Enter command (UPPER/LOWER/REVERSE/COUNT/VOWELS/DATE/TIME/QUIT): UPPER:hello world
Response: HELLO WORLD
Enter command (UPPER/LOWER/REVERSE/COUNT/VOWELS/DATE/TIME/QUIT): COUNT:network
Response: 7
Enter command (UPPER/LOWER/REVERSE/COUNT/VOWELS/DATE/TIME/QUIT): TIME
Response: 18:05:44.678674
Enter command (UPPER/LOWER/REVERSE/COUNT/VOWELS/DATE/TIME/QUIT): VOWELS:communication
Response: 6
Enter command (UPPER/LOWER/REVERSE/COUNT/VOWELS/DATE/TIME/QUIT): QUIT
Response: Disconnected
BUILD SUCCESSFUL (total time: 2 minutes 7 seconds)
|
```



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OUTPUTS:

Client 2:

```
NetLab2 (run) × NetLab2 (run) #2 × NetLab2 (run) #3 × NetLab2 (run) #4 ×
run:
client is ready
Enter command (UPPER/LOWER/REVERSE/COUNT/VOWELS/DATE/TIME/QUIT): LOWER:PROGRAMMING
Response: programming
Enter command (UPPER/LOWER/REVERSE/COUNT/VOWELS/DATE/TIME/QUIT): REVERSE:abcdef
Response: fedcba
Enter command (UPPER/LOWER/REVERSE/COUNT/VOWELS/DATE/TIME/QUIT): TIME
Response: 18:05:27.580497
Enter command (UPPER/LOWER/REVERSE/COUNT/VOWELS/DATE/TIME/QUIT): QUIT
Response: Disconnected
BUILD SUCCESSFUL (total time: 1 minute 15 seconds)
|
```

Client 3:

```
NetLab2 (run) × NetLab2 (run) #2 × NetLab2 (run) #3 × NetLab2 (run) #4 ×
run:
client is ready
Enter command (UPPER/LOWER/REVERSE/COUNT/VOWELS/DATE/TIME/QUIT): VOWELS:aeiouxyz
Response: 5
Enter command (UPPER/LOWER/REVERSE/COUNT/VOWELS/DATE/TIME/QUIT): COUNT:multithreading
Response: 14
Enter command (UPPER/LOWER/REVERSE/COUNT/VOWELS/DATE/TIME/QUIT): DATE
Response: 2026-04-07
Enter command (UPPER/LOWER/REVERSE/COUNT/VOWELS/DATE/TIME/QUIT): UNKNOWN:test
Response: ERROR: Unknown command from Client 3.
Enter command (UPPER/LOWER/REVERSE/COUNT/VOWELS/DATE/TIME/QUIT): QUIT
Response: Disconnected
BUILD SUCCESSFUL (total time: 1 minute 56 seconds)
|
```




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OUTPUTS:

server_log:

```
server_log.txt

[Time: 2026-04-07 18:04:15.748995] Client1 connected from 127.0.0.1:59037
[Time: 2026-04-07 18:04:19.266349] Client2 connected from 127.0.0.1:59038
[Time: 2026-04-07 18:04:22.179134] Client3 connected from 127.0.0.1:59039
[Time: 2026-04-07 18:04:35.777619] Client 1 | Request: UPPER:hello world | Response: HELLO WORLD
[Time: 2026-04-07 18:04:42.993878] Client 1 | Request: COUNT:network | Response: 7
[Time: 2026-04-07 18:04:55.199051] Client 2 | Request: LOWER:PROGRAMMING | Response:
programming
[Time: 2026-04-07 18:05:06.088712] Client 2 | Request: REVERSE:abcdef | Response: fedcba
[Time: 2026-04-07 18:05:15.009538] Client 3 | Request: VOWELS:aeiouxyz | Response: 5
[Time: 2026-04-07 18:05:22.421267] Client 3 | Request: COUNT:multithreading | Response: 14
[Time: 2026-04-07 18:05:27.580947] Client 2 | Request: TIME | Response: 18:05:27.580497
[Time: 2026-04-07 18:05:34.653351] Client 2 disconnected
[Time: 2026-04-07 18:05:44.679795] Client 1 | Request: TIME | Response: 18:05:44.678674
[Time: 2026-04-07 18:05:51.651691] Client 1 | Request: VOWELS:communication | Response: 6
[Time: 2026-04-07 18:06:00.686567] Client 3 | Request: DATE | Response: 2026-04-07
[Time: 2026-04-07 18:06:11.333512] Client 3 | Request: UNKNOWN:test | Response: ERROR: Unknown
command from Client 3.
[Time: 2026-04-07 18:06:18.512496] Client 3 disconnected
[Time: 2026-04-07 18:06:22.037279] Client 1 disconnected
```

Wireshark:

No.	Time	Source	Destination	Protocol	Length	Info
5	2026-04-07 18:04:14.357273	127.0.0.1	127.0.0.1	TCP	56	59035 → 57985 [FIN, ACK] Seq=1 Ack=1 Win=6379 Len=0 TSval=44183721 TSecr=2228406940
6	2026-04-07 18:04:14.357332	127.0.0.1	127.0.0.1	TCP	56	57985 → 59035 [ACK] Seq=1 Ack=2 Win=6371 Len=0 TSval=2228467938 TSecr=44183721
7	2026-04-07 18:04:14.359566	127.0.0.1	127.0.0.1	TCP	56	57985 → 59035 [FIN, ACK] Seq=1 Ack=2 Win=6371 Len=0 TSval=2228467940 TSecr=44183721
8	2026-04-07 18:04:14.359631	127.0.0.1	127.0.0.1	TCP	56	59035 → 57985 [ACK] Seq=2 Ack=2 Win=6379 Len=0 TSval=44183723 TSecr=2228467940
9	2026-04-07 18:04:15.725241	127.0.0.1	127.0.0.1	TCP	68	59037 → 4999 [SYN] Seq=0 Win=65535 Len=0 MSS=16344 WS=64 TSval=1051509729 TSecr=0 SA
10	2026-04-07 18:04:15.725368	127.0.0.1	127.0.0.1	TCP	68	4999 → 59037 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=16344 WS=64 TSval=976762406
11	2026-04-07 18:04:15.725393	127.0.0.1	127.0.0.1	TCP	56	59037 → 4999 [ACK] Seq=1 Ack=1 Win=408256 Len=0 TSval=1051509729 TSecr=976762406
12	2026-04-07 18:04:15.725411	127.0.0.1	127.0.0.1	TCP	56	[TCP Window Update] 4999 → 59037 [ACK] Seq=1 Ack=1 Win=408256 Len=0 TSval=976762406
13	2026-04-07 18:04:19.265815	127.0.0.1	127.0.0.1	TCP	68	59038 → 4999 [SYN] Seq=0 Win=65535 Len=0 MSS=16344 WS=64 TSval=3750322238 TSecr=0 SA
14	2026-04-07 18:04:19.265906	127.0.0.1	127.0.0.1	TCP	68	4999 → 59038 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=16344 WS=64 TSval=3140438364
15	2026-04-07 18:04:19.265926	127.0.0.1	127.0.0.1	TCP	56	59038 → 4999 [ACK] Seq=1 Ack=1 Win=408256 Len=0 TSval=3750322238 TSecr=3140438364
16	2026-04-07 18:04:19.265939	127.0.0.1	127.0.0.1	TCP	56	[TCP Window Update] 4999 → 59038 [ACK] Seq=1 Ack=1 Win=408256 Len=0 TSval=3140438364
17	2026-04-07 18:04:22.178625	127.0.0.1	127.0.0.1	TCP	68	59039 → 4999 [SYN] Seq=0 Win=65535 Len=0 MSS=16344 WS=64 TSval=3566321397 TSecr=0 SA
18	2026-04-07 18:04:22.178759	127.0.0.1	127.0.0.1	TCP	68	4999 → 59039 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=16344 WS=64 TSval=3408068738
19	2026-04-07 18:04:22.178785	127.0.0.1	127.0.0.1	TCP	56	59039 → 4999 [ACK] Seq=1 Ack=1 Win=408256 Len=0 TSval=3566321397 TSecr=3408068738
20	2026-04-07 18:04:22.178801	127.0.0.1	127.0.0.1	TCP	56	[TCP Window Update] 4999 → 59039 [ACK] Seq=1 Ack=1 Win=408256 Len=0 TSval=3408068738
21	2026-04-07 18:04:25.003708	127.0.0.1	127.0.0.1	TCP	68	59040 → 57985 [SYN] Seq=0 Win=65535 Len=0 MSS=16344 WS=64 TSval=983216366 TSecr=0 SA
22	2026-04-07 18:04:25.003765	127.0.0.1	127.0.0.1	TCP	68	57985 → 59040 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=16344 WS=64 TSval=4092227922
23	2026-04-07 18:04:25.003778	127.0.0.1	127.0.0.1	TCP	56	59040 → 57985 [ACK] Seq=1 Ack=1 Win=408256 Len=0 TSval=983216366 TSecr=4092227922
24	2026-04-07 18:04:25.003790	127.0.0.1	127.0.0.1	TCP	56	[TCP Window Update] 57985 → 59040 [ACK] Seq=1 Ack=1 Win=408256 Len=0 TSval=4092227922
25	2026-04-07 18:04:25.004253	127.0.0.1	127.0.0.1	TCP	520	59040 → 57985 [PSH, ACK] Seq=1 Ack=1 Win=408256 Len=464 TSval=983216367 TSecr=4092227922
26	2026-04-07 18:04:25.004320	127.0.0.1	127.0.0.1	TCP	56	57985 → 59040 [ACK] Seq=1 Ack=465 Win=407808 Len=0 TSval=4092227923 TSecr=983216367
27	2026-04-07 18:04:25.005852	127.0.0.1	127.0.0.1	HTTP/1.1	114	POST /notifications/libraries/sync HTTP/1.1, [JSON (application/json)]
28	2026-04-07 18:04:25.005888	127.0.0.1	127.0.0.1	TCP	56	57985 → 59040 [ACK] Seq=1 Ack=523 Win=407744 Len=0 TSval=4092227925 TSecr=983216369
29	2026-04-07 18:04:25.005922	127.0.0.1	127.0.0.1	TCP	74	59038 → 4999 [PSH, ACK] Seq=1 Ack=1 Win=408256 Len=18 TSval=1051529762 TSecr=976762406
30	2026-04-07 18:04:35.758391	127.0.0.1	127.0.0.1	TCP	56	4999 → 59037 [ACK] Seq=1 Ack=19 Win=408256 Len=0 TSval=976782439 TSecr=1051529762
31	2026-04-07 18:04:35.786737	127.0.0.1	127.0.0.1	TCP	68	4999 → 59037 [PSH, ACK] Seq=1 Ack=19 Win=408256 Len=12 TSval=976782468 TSecr=1051529762
32	2026-04-07 18:04:35.786801	127.0.0.1	127.0.0.1	TCP	56	59037 → 4999 [ACK] Seq=19 Ack=13 Win=408256 Len=0 TSval=1051529791 TSecr=976782468
33	2026-04-07 18:04:42.992053	127.0.0.1	127.0.0.1	TCP	70	59037 → 4999 [PSH, ACK] Seq=19 Ack=13 Win=408256 Len=14 TSval=1051536996 TSecr=976782468
34	2026-04-07 18:04:42.992109	127.0.0.1	127.0.0.1	TCP	56	4999 → 59037 [ACK] Seq=13 Ack=33 Win=408256 Len=0 TSval=976789673 TSecr=1051536996
35	2026-04-07 18:04:42.994643	127.0.0.1	127.0.0.1	TCP	58	4999 → 59037 [PSH, ACK] Seq=13 Ack=33 Win=408256 Len=2 TSval=976789676 TSecr=1051536996
36	2026-04-07 18:04:42.994710	127.0.0.1	127.0.0.1	TCP	56	59037 → 4999 [ACK] Seq=33 Ack=15 Win=408256 Len=0 TSval=1051536999 TSecr=976789676
37	2026-04-07 18:04:55.186914	127.0.0.1	127.0.0.1	TCP	74	59038 → 4999 [PSH, ACK] Seq=1 Ack=1 Win=408256 Len=18 TSval=3750322238 TSecr=3140438364
38	2026-04-07 18:04:55.186957	127.0.0.1	127.0.0.1	TCP	56	4999 → 59038 [ACK] Seq=1 Ack=19 Win=408256 Len=0 TSval=3140419585 TSecr=3750322238
39	2026-04-07 18:04:55.199728	127.0.0.1	127.0.0.1	TCP	68	4999 → 59038 [PSH, ACK] Seq=1 Ack=19 Win=408256 Len=12 TSval=3140419598 TSecr=3750322238
40	2026-04-07 18:04:55.199777	127.0.0.1	127.0.0.1	TCP	56	59038 → 4999 [ACK] Seq=19 Ack=13 Win=408256 Len=0 TSval=3750322251 TSecr=3140419598
41	2026-04-07 18:05:06.075182	127.0.0.1	127.0.0.1	TCP	71	59038 → 4999 [PSH, ACK] Seq=19 Ack=13 Win=408256 Len=15 TSval=375033127 TSecr=3140419598
42	2026-04-07 18:05:06.075233	127.0.0.1	127.0.0.1	TCP	56	4999 → 59038 [ACK] Seq=13 Ack=34 Win=408256 Len=0 TSval=3140430474 TSecr=375033127
43	2026-04-07 18:05:06.089467	127.0.0.1	127.0.0.1	TCP	63	4999 → 59038 [PSH, ACK] Seq=13 Ack=34 Win=408256 Len=7 TSval=3140430488 TSecr=375033127
44	2026-04-07 18:05:06.089522	127.0.0.1	127.0.0.1	TCP	56	59038 → 4999 [ACK] Seq=34 Ack=20 Win=408256 Len=0 TSval=3750333141 TSecr=3140430488
45	2026-04-07 18:05:15.001165	127.0.0.1	127.0.0.1	TCP	72	59039 → 4999 [PSH, ACK] Seq=1 Ack=1 Win=408256 Len=16 TSval=3566374221 TSecr=3408068738
46	2026-04-07 18:05:15.001229	127.0.0.1	127.0.0.1	TCP	56	4999 → 59039 [ACK] Seq=1 Ack=17 Win=408256 Len=0 TSval=3408121562 TSecr=3566374221
47	2026-04-07 18:05:15.010498	127.0.0.1	127.0.0.1	TCP	58	4999 → 59039 [PSH, ACK] Seq=1 Ack=17 Win=408256 Len=2 TSval=3408121571 TSecr=3566374221



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OUTPUTS:

Wireshark Cont.:

No.	Time	Source	Destination	Protocol	Length	Info
43	2026-04-07 18:05:06.089467	127.0.0.1	127.0.0.1	TCP	63	4999 → 59038 [PSH, ACK] Seq=13 Ack=34 Win=408256 Len=7 TSval=3140430488 TSecr=3750331
44	2026-04-07 18:05:06.089522	127.0.0.1	127.0.0.1	TCP	56	59038 → 4999 [ACK] Seq=34 Ack=20 Win=408256 Len=0 TSval=3750333141 TSecr=3140430488
45	2026-04-07 18:05:15.001165	127.0.0.1	127.0.0.1	TCP	72	59039 → 4999 [PSH, ACK] Seq=1 Ack=1 Win=408256 Len=16 TSval=3566374221 TSecr=3408066
46	2026-04-07 18:05:15.001229	127.0.0.1	127.0.0.1	TCP	56	4999 → 59039 [ACK] Seq=1 Ack=17 Win=408256 Len=0 TSval=3408121562 TSecr=3566374221
47	2026-04-07 18:05:15.010498	127.0.0.1	127.0.0.1	TCP	58	4999 → 59039 [PSH, ACK] Seq=17 Ack=17 Win=408256 Len=2 TSval=3408121571 TSecr=3566374
48	2026-04-07 18:05:15.010538	127.0.0.1	127.0.0.1	TCP	56	59039 → 4999 [ACK] Seq=17 Ack=3 Win=408256 Len=0 TSval=3566374230 TSecr=3408121571
53	2026-04-07 18:05:22.416939	127.0.0.1	127.0.0.1	TCP	77	59039 → 4999 [PSH, ACK] Seq=17 Ack=3 Win=408256 Len=21 TSval=3566381637 TSecr=340812
54	2026-04-07 18:05:22.416991	127.0.0.1	127.0.0.1	TCP	56	4999 → 59039 [ACK] Seq=3 Ack=38 Win=408256 Len=0 TSval=3408128078 TSecr=3566381637
55	2026-04-07 18:05:22.422173	127.0.0.1	127.0.0.1	TCP	56	4999 → 59039 [PSH, ACK] Seq=3 Ack=38 Win=408256 Len=3 TSval=3408128083 TSecr=3566381
56	2026-04-07 18:05:22.422229	127.0.0.1	127.0.0.1	TCP	56	59039 → 4999 [ACK] Seq=38 Ack=38 Win=408256 Len=0 TSval=3566381642 TSecr=3408128093
57	2026-04-07 18:05:26.008364	127.0.0.1	127.0.0.1	TCP	56	59040 → 57985 [FIN, ACK] Seq=523 Ack=1 Win=408256 Len=0 TSval=983277372 TSecr=409222
58	2026-04-07 18:05:26.008441	127.0.0.1	127.0.0.1	TCP	56	57985 → 59040 [ACK] Seq=1 Ack=524 Win=407744 Len=0 TSval=4092288928 TSecr=983277372
59	2026-04-07 18:05:26.010561	127.0.0.1	127.0.0.1	TCP	56	57985 → 59040 [FIN, ACK] Seq=1 Ack=524 Win=407744 Len=0 TSval=4092288930 TSecr=98327
60	2026-04-07 18:05:26.010622	127.0.0.1	127.0.0.1	TCP	56	59040 → 57985 [ACK] Seq=524 Ack=2 Win=408256 Len=0 TSval=9832773734 TSecr=4092288930
61	2026-04-07 18:05:26.025500	127.0.0.1	127.0.0.1	TCP	68	59043 → 57985 [SYN] Seq=0 Win=65535 Len=0 MSS=16344 WS=64 TSval=1461718160 TSecr=0
62	2026-04-07 18:05:26.025560	127.0.0.1	127.0.0.1	TCP	68	57985 → 59043 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=16344 WS=64 TSval=278270647 TSecr=0
63	2026-04-07 18:05:26.025572	127.0.0.1	127.0.0.1	TCP	56	59043 → 57985 [ACK] Seq=1 Ack=1 Win=408256 Len=0 TSval=1461718162 TSecr=2782706470
64	2026-04-07 18:05:26.025583	127.0.0.1	127.0.0.1	TCP	56	[TCP Window Update] 57985 → 59043 [ACK] Seq=1 Ack=1 Win=408256 Len=0 TSval=278270647
65	2026-04-07 18:05:26.026974	127.0.0.1	127.0.0.1	TCP	520	59043 → 57985 [PSH, ACK] Seq=1 Ack=1 Win=408256 Len=464 TSval=1461718163 TSecr=27827
66	2026-04-07 18:05:26.027019	127.0.0.1	127.0.0.1	TCP	56	57985 → 59043 [ACK] Seq=1 Ack=465 Win=407888 Len=0 TSval=2782706471 TSecr=1461718163
67	2026-04-07 18:05:26.027539	127.0.0.1	127.0.0.1	HTTP...	114	POST /notifications/libraries/sync HTTP/1.1, JSON (application/json)
68	2026-04-07 18:05:26.027562	127.0.0.1	127.0.0.1	TCP	56	57985 → 59043 [ACK] Seq=1 Ack=523 Win=407744 Len=0 TSval=2782706472 TSecr=1461718163
69	2026-04-07 18:05:27.580220	127.0.0.1	127.0.0.1	TCP	61	59038 → 4999 [PSH, ACK] Seq=34 Ack=20 Win=408256 Len=5 TSval=3750354632 TSecr=314043
70	2026-04-07 18:05:27.580258	127.0.0.1	127.0.0.1	TCP	49	4999 → 59038 [ACK] Seq=20 Ack=39 Win=408256 Len=0 TSval=3140451979 TSecr=3750354632
71	2026-04-07 18:05:27.581483	127.0.0.1	127.0.0.1	TCP	72	4999 → 59038 [PSH, ACK] Seq=20 Ack=39 Win=408256 Len=16 TSval=3140451981 TSecr=375035
72	2026-04-07 18:05:27.581544	127.0.0.1	127.0.0.1	TCP	56	59038 → 4999 [ACK] Seq=39 Ack=36 Win=408256 Len=0 TSval=3750354634 TSecr=3140451981
73	2026-04-07 18:05:34.645021	127.0.0.1	127.0.0.1	TCP	61	59038 → 4999 [PSH, ACK] Seq=39 Ack=36 Win=408256 Len=5 TSval=3750361697 TSecr=314045
74	2026-04-07 18:05:34.645064	127.0.0.1	127.0.0.1	TCP	49	4999 → 59038 [ACK] Seq=36 Ack=44 Win=408256 Len=0 TSval=3140459044 TSecr=3750361697
75	2026-04-07 18:05:34.652763	127.0.0.1	127.0.0.1	TCP	69	4999 → 59038 [PSH, ACK] Seq=36 Ack=44 Win=408256 Len=13 TSval=3140459052 TSecr=37503
76	2026-04-07 18:05:34.652835	127.0.0.1	127.0.0.1	TCP	56	59038 → 4999 [ACK] Seq=44 Ack=49 Win=408192 Len=0 TSval=3750361705 TSecr=3140459052
77	2026-04-07 18:05:34.650864	127.0.0.1	127.0.0.1	TCP	56	59038 → 4999 [FIN, ACK] Seq=44 Ack=49 Win=408192 Len=0 TSval=3750361710 TSecr=314045
78	2026-04-07 18:05:34.658114	127.0.0.1	127.0.0.1	TCP	56	4999 → 59038 [ACK] Seq=49 Ack=45 Win=408256 Len=0 TSval=3140459057 TSecr=3750361710
79	2026-04-07 18:05:34.663595	127.0.0.1	127.0.0.1	TCP	56	4999 → 59038 [FIN, ACK] Seq=49 Ack=45 Win=408256 Len=0 TSval=3140459063 TSecr=37503
80	2026-04-07 18:05:34.663664	127.0.0.1	127.0.0.1	TCP	56	59038 → 4999 [ACK] Seq=45 Ack=50 Win=408192 Len=0 TSval=3750361716 TSecr=3140459063
81	2026-04-07 18:05:44.677690	127.0.0.1	127.0.0.1	TCP	61	59037 → 4999 [PSH, ACK] Seq=33 Ack=15 Win=408256 Len=5 TSval=1051598683 TSecr=97687
82	2026-04-07 18:05:44.677745	127.0.0.1	127.0.0.1	TCP	56	4999 → 59037 [ACK] Seq=15 Ack=38 Win=408256 Len=0 TSval=976851360 TSecr=1051598683
83	2026-04-07 18:05:44.680296	127.0.0.1	127.0.0.1	TCP	72	4999 → 59037 [PSH, ACK] Seq=15 Ack=38 Win=408256 Len=16 TSval=976851363 TSecr=105159
84	2026-04-07 18:05:44.680338	127.0.0.1	127.0.0.1	TCP	56	59037 → 4999 [PSH, ACK] Seq=38 Ack=31 Win=408256 Len=21 TSval=1051605656 TSecr=97685
85	2026-04-07 18:05:44.680338	127.0.0.1	127.0.0.1	TCP	56	59037 → 4999 [ACK] Seq=38 Ack=31 Win=408256 Len=0 TSval=1051598686 TSecr=976851363
86	2026-04-07 18:05:51.650297	127.0.0.1	127.0.0.1	TCP	77	59037 → 4999 [PSH, ACK] Seq=38 Ack=31 Win=408256 Len=21 TSval=1051605656 TSecr=97685
87	2026-04-07 18:05:51.652310	127.0.0.1	127.0.0.1	TCP	56	4999 → 59037 [ACK] Seq=31 Ack=59 Win=408192 Len=0 TSval=976858333 TSecr=1051605656
88	2026-04-07 18:05:51.652310	127.0.0.1	127.0.0.1	TCP	58	4999 → 59037 [PSH, ACK] Seq=31 Ack=59 Win=408192 Len=2 TSval=976858335 TSecr=1051605
89	2026-04-07 18:05:51.652345	127.0.0.1	127.0.0.1	TCP	56	59037 → 4999 [ACK] Seq=59 Ack=33 Win=408256 Len=0 TSval=1051605658 TSecr=976858335
89	2026-04-07 18:06:00.685141	127.0.0.1	127.0.0.1	TCP	61	59039 → 4999 [PSH, ACK] Seq=38 Ack=6 Win=408256 Len=5 TSval=3566419906 TSecr=3408128
89	2026-04-07 18:05:51.652310	127.0.0.1	127.0.0.1	TCP	56	4999 → 59037 [FIN, ACK] Seq=68 Ack=61 Win=408192 Len=0 TSval=3408185076 TSecr=356643
90	2026-04-07 18:06:00.685141	127.0.0.1	127.0.0.1	TCP	56	59039 → 4999 [ACK] Seq=38 Ack=6 Win=408256 Len=5 TSval=3566419906 TSecr=3408128
91	2026-04-07 18:06:00.687167	127.0.0.1	127.0.0.1	TCP	67	4999 → 59039 [PSH, ACK] Seq=6 Ack=43 Win=408256 Len=11 TSval=3408167249 TSecr=356641
92	2026-04-07 18:06:00.687221	127.0.0.1	127.0.0.1	TCP	56	59039 → 4999 [ACK] Seq=43 Ack=17 Win=408256 Len=0 TSval=3566419908 TSecr=3408167249
93	2026-04-07 18:06:11.331772	127.0.0.1	127.0.0.1	TCP	69	59039 → 4999 [PSH, ACK] Seq=43 Ack=17 Win=408256 Len=13 TSval=3566430553 TSecr=34081
94	2026-04-07 18:06:11.331818	127.0.0.1	127.0.0.1	TCP	49	4999 → 59039 [ACK] Seq=17 Ack=56 Win=408192 Len=0 TSval=3408177894 TSecr=3566430553
95	2026-04-07 18:06:11.331844	127.0.0.1	127.0.0.1	TCP	94	4999 → 59039 [PSH, ACK] Seq=17 Ack=56 Win=408192 Len=38 TSval=3408177896 TSecr=35664
96	2026-04-07 18:06:11.334137	127.0.0.1	127.0.0.1	TCP	56	59039 → 4999 [ACK] Seq=56 Ack=55 Win=408192 Len=0 TSval=3566430555 TSecr=3408177896
97	2026-04-07 18:06:11.511194	127.0.0.1	127.0.0.1	TCP	61	59039 → 4999 [PSH, ACK] Seq=56 Ack=55 Win=408192 Len=5 TSval=3566437732 TSecr=340817
98	2026-04-07 18:06:11.511238	127.0.0.1	127.0.0.1	TCP	49	4999 → 59039 [ACK] Seq=55 Ack=61 Win=408192 Len=0 TSval=3408185073 TSecr=3566437732
99	2026-04-07 18:06:11.511996	127.0.0.1	127.0.0.1	TCP	69	4999 → 59039 [PSH, ACK] Seq=55 Ack=61 Win=408192 Len=13 TSval=3408185073 TSecr=35664
100	2026-04-07 18:06:18.512030	127.0.0.1	127.0.0.1	TCP	56	59039 → 4999 [ACK] Seq=61 Ack=68 Win=408192 Len=0 TSval=3566437732 TSecr=3408185073
101	2026-04-07 18:06:18.513786	127.0.0.1	127.0.0.1	TCP	49	4999 → 59039 [FIN, ACK] Seq=68 Ack=61 Win=408192 Len=0 TSval=3408185076 TSecr=356643
102	2026-04-07 18:06:18.513864	127.0.0.1	127.0.0.1	TCP	56	59039 → 4999 [ACK] Seq=61 Ack=69 Win=408192 Len=0 TSval=3566437735 TSecr=3408185076
103	2026-04-07 18:06:18.517655	127.0.0.1	127.0.0.1	TCP	56	59039 → 4999 [FIN, ACK] Seq=61 Ack=69 Win=408192 Len=0 TSval=3566437739 TSecr=340818
104	2026-04-07 18:06:18.517725	127.0.0.1	127.0.0.1	TCP	56	4999 → 59039 [ACK] Seq=69 Ack=62 Win=408192 Len=0 TSval=3408185098 TSecr=3566437739
105	2026-04-07 18:06:22.036681	127.0.0.1	127.0.0.1	TCP	61	59037 → 4999 [PSH, ACK] Seq=59 Ack=33 Win=408256 Len=5 TSval=1051636043 TSecr=97685
106	2026-04-07 18:06:22.036684	127.0.0.1	127.0.0.1	TCP	56	4999 → 59037 [ACK] Seq=33 Ack=64 Win=408192 Len=0 TSval=976888720 TSecr=1051636043
107	2026-04-07 18:06:22.036979	127.0.0.1	127.0.0.1	TCP	49	4999 → 59037 [PSH, ACK] Seq=33 Ack=64 Win=408192 Len=13 TSval=976888720 TSecr=105163
108	2026-04-07 18:06:22.036997	127.0.0.1	127.0.0.1	TCP	56	59037 → 4999 [ACK] Seq=64 Ack=46 Win=408192 Len=0 TSval=1051636043 TSecr=976888720
109	2026-04-07 18:06:22.037921	127.0.0.1	127.0.0.1	TCP	49	4999 → 59037 [FIN, ACK] Seq=46 Ack=64 Win=408192 Len=0 TSval=976888721 TSecr=105163
110	2026-04-07 18:06:22.043758	127.0.0.1	127.0.0.1	TCP	56	59037 → 4999 [ACK] Seq=64 Ack=47 Win=408192 Len=0 TSval=1051636044 TSecr=976888721
111	2026-04-07 18:06:22.041392	127.0.0.1	127.0.0.1	TCP	56	59037 → 4999 [FIN, ACK] Seq=64 Ack=47 Win=408192 Len=0 TSval=1051636047 TSecr=976888
112	2026-04-07 18:06:22.041486	127.0.0.1	127.0.0.1	TCP	56	4999 → 59037 [ACK] Seq=47 Ack=65 Win=408192 Len=0 TSval=976888721 TSecr=1051636047
113	2026-04-07 18:06:22.023672	127.0.0.1	127.0.0.1	TCP	56	59043 → 57985 [FIN, ACK] Seq=523 Ack=1 Win=408256 Len=0 TSval=1461779161 TSecr=27827
114	2026-04-07 18:06:27.023731	127.0.0.1	127.0.0.1	TCP	56	57985 → 59043 [ACK] Seq=1 Ack=524 Win=407744 Len=0 TSval=2782767469 TSecr=1461779161
115	2026-04-07 18:06:27.025810	127.0.0.1	127.0.0.1	TCP	56	57985 → 59043 [FIN, ACK] Seq=1 Ack=524 Win=407744 Len=0 TSval=2782767470 TSecr=14617
116	2026-04-07 18:06:27.025896	127.0.0.1	127.0.0.1	TCP	56	59043 → 57985 [ACK] Seq=524 Ack=2 Win=408256 Len=0 TSval=1461779163 TSecr=2782767470
117	2026-04-07 18:06:28.004210	127.0.0.1	127.0.0.1	TCP	68	59044 → 57985 [SYN] Seq=0 Win=65535 Len=0 MSS=16344 WS=64 TSval=1952142402 TSecr=0
118	2026-04-07 18:06:28.004278	127.0.0.1	127.0.0.1	TCP	68	57985 → 59044 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=16344 WS=64 TSval=1952142402 TSecr=0
119	2026-04-07 18:06:28.004293	127.0.0.1	127.0.0.1	TCP	56	59044 → 57985 [ACK] Seq=1 Ack=1 Win=408256 Len=0 TSval=2782767470 TSecr=1952142402
120	2026-04-07 18:06:28.004310	127.0.0.1	127.0.0.1	TCP	56	[TCP Window Update] 57985 → 59044 [ACK] Seq=1 Ack=1 Win=408256 Len=0 TSval=1952142402 TSecr=0
121	2026-04-07 18:06:28.004883	127.0.0.1	127.0.0.1	TCP	520	59044 → 57985 [PSH, ACK] Seq=1 Ack=1 Win=408256 Len=464 TSval=2782767470 TSecr=19521
122	2026-04-07 18:06:28.004999	127.0.0.1	127.0.0.1	TCP	56	57985 → 59044 [ACK] Seq=1 Ack=465 Win=407888 Len=0 TSval=1952142402 TSecr=2782767470
123	2026-04-07 18:06:28.007359	127.0.0.1	127.0.0.1	HTTP...	114	POST /notifications/libraries/sync HTTP/1.1, JSON (application/json)
124	2026-04-07 18:06:28.007392	127.0.0.1	127.0.0.1	TCP	56	57985 → 59044 [ACK] Seq=1 Ack=523 Win=407744 Len=0 TSval=1952142405 TSecr=2782767470



COMPUTER NETWORKS – SPRING 2026

OUTPUTS:

Wireshark Cont.:

Client 3: COUNT:multithreading

Request:

Wireshark packet capture showing a request from Client 3. The packet list shows a POST request to /notifications/libraries/sync. The packet details show the request body as a JSON object. The packet bytes show the raw data.

No.	Time	Source	Destination	Protocol	Length	Info
53	2026-04-07 18:05:22.416939	127.0.0.1	127.0.0.1	TCP	77	59039 → 4999 [PSH, ACK] Seq=17 Ack=3 Win=488256 Len=21 TSval=3566381637 TSecr=3408128978
54	2026-04-07 18:05:22.416991	127.0.0.1	127.0.0.1	TCP	56	4999 → 59039 [ACK] Seq=3 Ack=38 Win=488256 Len=0 TSval=3408128978 TSecr=3566381637
55	2026-04-07 18:05:22.422173	127.0.0.1	127.0.0.1	TCP	59	4999 → 59039 [PSH, ACK] Seq=3 Ack=38 Win=488256 Len=3 TSval=3408128983 TSecr=3566381637
56	2026-04-07 18:05:22.422229	127.0.0.1	127.0.0.1	TCP	56	59039 → 4999 [ACK] Seq=38 Ack=6 Win=488256 Len=0 TSval=3566381642 TSecr=3408128983
57	2026-04-07 18:05:26.008364	127.0.0.1	127.0.0.1	TCP	56	59040 → 57985 [FIN, ACK] Seq=523 Ack=1 Win=488256 Len=0 TSval=983277372 TSecr=4892289930
58	2026-04-07 18:05:26.008441	127.0.0.1	127.0.0.1	TCP	56	57985 → 59040 [ACK] Seq=1 Ack=524 Win=488256 Len=0 TSval=4892289928 TSecr=983277372
59	2026-04-07 18:05:26.010561	127.0.0.1	127.0.0.1	TCP	56	57985 → 59040 [FIN, ACK] Seq=1 Ack=524 Win=488256 Len=0 TSval=4892289930 TSecr=983277372
60	2026-04-07 18:05:26.010622	127.0.0.1	127.0.0.1	TCP	56	59040 → 57985 [ACK] Seq=524 Ack=2 Win=488256 Len=0 TSval=983277374 TSecr=4892289930
61	2026-04-07 18:05:26.025500	127.0.0.1	127.0.0.1	TCP	68	59043 → 57985 [SYN] Seq=0 Win=65535 Len=0 MSS=16344 WS=64 TSval=1461718160 TSecr=0
62	2026-04-07 18:05:26.025560	127.0.0.1	127.0.0.1	TCP	68	57985 → 59043 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=16344 WS=64 TSval=2782706470 TSecr=0
63	2026-04-07 18:05:26.025572	127.0.0.1	127.0.0.1	TCP	56	59043 → 57985 [ACK] Seq=1 Ack=1 Win=488256 Len=0 TSval=1461718162 TSecr=2782706470
64	2026-04-07 18:05:26.025583	127.0.0.1	127.0.0.1	TCP	56	[TCP Window Update] 57985 → 59043 [ACK] Seq=1 Ack=1 Win=488256 Len=0 TSval=2782706470 TSecr=2782706470
65	2026-04-07 18:05:26.026974	127.0.0.1	127.0.0.1	TCP	520	59043 → 57985 [PSH, ACK] Seq=1 Ack=1 Win=488256 Len=464 TSval=1461718163 TSecr=2782706470
66	2026-04-07 18:05:26.027019	127.0.0.1	127.0.0.1	TCP	56	57985 → 59043 [ACK] Seq=1 Ack=465 Win=488256 Len=0 TSval=2782706471 TSecr=1461718163
67	2026-04-07 18:05:26.027539	127.0.0.1	127.0.0.1	HTTP	114	POST /notifications/libraries/sync HTTP/1.1, JSON (application/json)
68	2026-04-07 18:05:26.027562	127.0.0.1	127.0.0.1	TCP	56	57985 → 59043 [ACK] Seq=1 Ack=523 Win=488256 Len=0 TSval=2782706472 TSecr=1461718163
69	2026-04-07 18:05:27.580228	127.0.0.1	127.0.0.1	TCP	61	59038 → 4999 [PSH, ACK] Seq=34 Ack=20 Win=488256 Len=5 TSval=3758354632 TSecr=314043
70	2026-04-07 18:05:27.580258	127.0.0.1	127.0.0.1	TCP	56	4999 → 59038 [ACK] Seq=20 Ack=39 Win=488256 Len=0 TSval=3140451979 TSecr=3758354632
71	2026-04-07 18:05:27.581483	127.0.0.1	127.0.0.1	TCP	72	4999 → 59038 [PSH, ACK] Seq=20 Ack=39 Win=488256 Len=16 TSval=3140451981 TSecr=3758354632
72	2026-04-07 18:05:27.581514	127.0.0.1	127.0.0.1	TCP	56	59038 → 4999 [ACK] Seq=39 Ack=36 Win=488256 Len=0 TSval=3758354634 TSecr=3140451981
73	2026-04-07 18:05:34.645821	127.0.0.1	127.0.0.1	TCP	61	59038 → 4999 [PSH, ACK] Seq=39 Ack=36 Win=488256 Len=5 TSval=3758361697 TSecr=3140451981
74	2026-04-07 18:05:34.645864	127.0.0.1	127.0.0.1	TCP	56	4999 → 59038 [ACK] Seq=36 Ack=44 Win=488256 Len=0 TSval=3140459044 TSecr=3758361697
75	2026-04-07 18:05:34.652763	127.0.0.1	127.0.0.1	TCP	69	4999 → 59038 [PSH, ACK] Seq=36 Ack=44 Win=488256 Len=13 TSval=3140459052 TSecr=3758361697

Frame 53: Packet, 77 bytes on wire (616 bits), 77 bytes captured (616 bits) on interface
Null/Loopback
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 59039, Dst Port: 4999, Seq: 17, Ack: 3, Len: 21
Data (21 bytes)

Response:

Wireshark packet capture showing a response from Client 3. The packet list shows a POST request to /notifications/libraries/sync. The packet details show the request body as a JSON object. The packet bytes show the raw data.

No.	Time	Source	Destination	Protocol	Length	Info
48	2026-04-07 18:05:15.010538	127.0.0.1	127.0.0.1	TCP	56	59039 → 4999 [ACK] Seq=17 Ack=3 Win=488256 Len=0 TSval=3566381637 TSecr=3408128978
53	2026-04-07 18:05:22.416939	127.0.0.1	127.0.0.1	TCP	77	59039 → 4999 [PSH, ACK] Seq=17 Ack=3 Win=488256 Len=21 TSval=3566381637 TSecr=3408128978
54	2026-04-07 18:05:22.416991	127.0.0.1	127.0.0.1	TCP	56	4999 → 59039 [ACK] Seq=3 Ack=38 Win=488256 Len=0 TSval=3408128978 TSecr=3566381637
55	2026-04-07 18:05:22.422173	127.0.0.1	127.0.0.1	TCP	59	4999 → 59039 [PSH, ACK] Seq=3 Ack=38 Win=488256 Len=3 TSval=3408128983 TSecr=3566381637
56	2026-04-07 18:05:22.422229	127.0.0.1	127.0.0.1	TCP	56	59039 → 4999 [ACK] Seq=38 Ack=6 Win=488256 Len=0 TSval=3566381642 TSecr=3408128983
57	2026-04-07 18:05:26.008364	127.0.0.1	127.0.0.1	TCP	56	59040 → 57985 [FIN, ACK] Seq=523 Ack=1 Win=488256 Len=0 TSval=983277372 TSecr=4892289930
58	2026-04-07 18:05:26.008441	127.0.0.1	127.0.0.1	TCP	56	57985 → 59040 [ACK] Seq=1 Ack=524 Win=488256 Len=0 TSval=4892289928 TSecr=983277372
59	2026-04-07 18:05:26.010561	127.0.0.1	127.0.0.1	TCP	56	57985 → 59040 [FIN, ACK] Seq=1 Ack=524 Win=488256 Len=0 TSval=4892289930 TSecr=983277372
60	2026-04-07 18:05:26.010622	127.0.0.1	127.0.0.1	TCP	56	59040 → 57985 [ACK] Seq=524 Ack=2 Win=488256 Len=0 TSval=983277374 TSecr=4892289930
61	2026-04-07 18:05:26.025500	127.0.0.1	127.0.0.1	TCP	68	59043 → 57985 [SYN] Seq=0 Win=65535 Len=0 MSS=16344 WS=64 TSval=1461718160 TSecr=0
62	2026-04-07 18:05:26.025560	127.0.0.1	127.0.0.1	TCP	68	57985 → 59043 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=16344 WS=64 TSval=2782706470 TSecr=0
63	2026-04-07 18:05:26.025572	127.0.0.1	127.0.0.1	TCP	56	59043 → 57985 [ACK] Seq=1 Ack=1 Win=488256 Len=0 TSval=1461718162 TSecr=2782706470
64	2026-04-07 18:05:26.025583	127.0.0.1	127.0.0.1	TCP	56	[TCP Window Update] 57985 → 59043 [ACK] Seq=1 Ack=1 Win=488256 Len=0 TSval=2782706470 TSecr=2782706470
65	2026-04-07 18:05:26.026974	127.0.0.1	127.0.0.1	TCP	520	59043 → 57985 [PSH, ACK] Seq=1 Ack=1 Win=488256 Len=464 TSval=1461718163 TSecr=2782706470
66	2026-04-07 18:05:26.027019	127.0.0.1	127.0.0.1	TCP	56	57985 → 59043 [ACK] Seq=1 Ack=465 Win=488256 Len=0 TSval=2782706471 TSecr=1461718163
67	2026-04-07 18:05:26.027562	127.0.0.1	127.0.0.1	HTTP	114	POST /notifications/libraries/sync HTTP/1.1, JSON (application/json)
68	2026-04-07 18:05:26.027562	127.0.0.1	127.0.0.1	TCP	56	57985 → 59043 [ACK] Seq=1 Ack=523 Win=488256 Len=0 TSval=2782706472 TSecr=1461718163
69	2026-04-07 18:05:27.580228	127.0.0.1	127.0.0.1	TCP	61	59038 → 4999 [PSH, ACK] Seq=34 Ack=20 Win=488256 Len=5 TSval=3758354632 TSecr=314043
70	2026-04-07 18:05:27.580258	127.0.0.1	127.0.0.1	TCP	56	4999 → 59038 [ACK] Seq=20 Ack=39 Win=488256 Len=0 TSval=3140451979 TSecr=3758354632
71	2026-04-07 18:05:27.581483	127.0.0.1	127.0.0.1	TCP	72	4999 → 59038 [PSH, ACK] Seq=20 Ack=39 Win=488256 Len=16 TSval=3140451981 TSecr=3758354632
72	2026-04-07 18:05:27.581514	127.0.0.1	127.0.0.1	TCP	56	59038 → 4999 [ACK] Seq=39 Ack=36 Win=488256 Len=0 TSval=3758354634 TSecr=3140451981
73	2026-04-07 18:05:34.645821	127.0.0.1	127.0.0.1	TCP	61	59038 → 4999 [PSH, ACK] Seq=39 Ack=36 Win=488256 Len=5 TSval=3758361697 TSecr=3140451981
74	2026-04-07 18:05:34.645864	127.0.0.1	127.0.0.1	TCP	56	4999 → 59038 [ACK] Seq=36 Ack=44 Win=488256 Len=0 TSval=3140459044 TSecr=3758361697
75	2026-04-07 18:05:34.652763	127.0.0.1	127.0.0.1	TCP	69	4999 → 59038 [PSH, ACK] Seq=36 Ack=44 Win=488256 Len=13 TSval=3140459052 TSecr=3758361697

Frame 55: Packet, 59 bytes on wire (472 bits), 59 bytes captured (472 bits) on interface
Null/Loopback
Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
Transmission Control Protocol, Src Port: 4999, Dst Port: 59039, Seq: 3, Ack: 38, Len: 3
Data (3 bytes)



THANK YOU.